

PROFIT MAXIMIZATION, INDUSTRY SUPPLY, AND EQUILIBRIUM IN PERFECTLY COMPETITIVE MARKETS PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

Medina just opened a coffee shop in NU named “Steely Coffee”. Hoping to cater to the NU Finance Bros, she sells a special kind of brew infused with nitrogen, which she dubs ‘nitro coffee’. Assume for the purposes of this question, that nitro coffee requires coffee beans and a machine to add nitrogen and transform it in liquid form. The coffee beans come in half-kilogram sacks, c , and she rents time on a machine by the hour, m . The price of coffee beans is p_c , the price of machine hours is p_m . She sells her nitro coffee, q by the cup (only one size).

- (a) After producing for a while, Medina knows that there is not much flexibility in the production process. The machine takes half an hour to transform 500 grams of coffee beans into a cup of nitro coffee. Running the machine longer doesn’t make more without extra coffee beans, and extra coffee beans (if not allowed enough time in the machine) doesn’t get enough nitrogen to produce nitro coffee. Which of the following production functions explain this relationship?
- (i) $f_1(m, c) = m^a c^b$,
 - (ii) $f_2(m, c) = am + bc$,
 - (iii) $f_3(m, c) = \min\{am, bc\}$.

What are the values of the parameters a and b for the production function that you chose?¹

- (b) Medina had previously signed an agreement to rent the machine for exactly 8 hours per day (one machine, full time). Assuming that she can’t get out of the contract she’s already committed to, and that her shop only has room for one machine, find Medina’s fixed, variable, average variable, and marginal costs (as a function of the prices of inputs and the number of cups of nitro coffee she produces).
- (c) Find the price at which Medina would shut-down assuming that she has already paid the machine rental.
- (d) Find the price at which Medina would shut-down assuming her contract is up and she’s deciding whether or not to rent the machine again (using the same contract of 8 hours per day).

Problem 2

In this exercise, you will explore the equilibrium of a market in which we know the cost function of the firms, as well as the demand function of consumers. You will first analyze the short-run equilibrium,

¹Remember that we are measuring output (nitro coffee) in cups, and inputs in hours of machine and half-kilograms of coffee means.

and then the long-run equilibrium. Each firm in the industry has the same technology with cost functions

$$c_{SR}(y) = k^2 + 2y + y^2$$

$$c_{LR}(y) = \begin{cases} 0 & \text{if } y = 0, \\ k^2 + 2y + y^2 & \text{if } y > 0. \end{cases}$$

where y is the quantity produced and k is some positive fixed cost. Notice that the short-run and long-run cost functions are the same except when $y = 0$. Although *there are fixed costs in both the short run and long run (remember that this is possible)*, in the short-run they are *unavoidable* while in the long-run the firm *can avoid them* by not producing. You should think of this cost function as coming out of a standard cost minimization problem with respect to the inputs.

- Derive the average cost, the average variable cost and the marginal cost. Graph all these cost curves in an appropriately labeled diagram.
- Draw the supply function in the graph, and write down the equation that represents it, $y(p)$.
- Derive the aggregate supply function $Y^S(p)$ by adding up the supply $y(p)$ over the J firms that are in the market. Write down the expression for $Y^S(p)$.
- Consider now the demand side of the market. For simplicity, assume a linear demand function: $Y^D(p) = A - Bp$, where A and B are positive numbers, and Y^D represents the total quantity demanded in the industry. Assume that this comes from aggregation of the individual demand functions derived from maximization. Find the short-run equilibrium price p^* by equating Y^D and Y^S . You can assume that A and B are such that we are on the increasing part of the supply function, i.e. quantity produced is strictly positive. Find the short-run equilibrium industry production $Y^* = Y^D(p^*) = Y^S(p^*)$.
- Under what conditions for A , B , J and k the firms will indeed produce a positive quantity of output? We maintain this assumption for the next items.
- In the short-run, what happens to Y^* and p^* as the fixed cost k increases? What is the intuition?
- Consider now the long-run equilibrium, in which firms are allowed to enter the market. Solve for the number of firms J^* that will enter into the market.
- How does the number of firms J^* depend on k , A and B ?
- What is the equation of the long-run supply curve? Is it horizontal, increasing or decreasing?

Problem 3

Suppose that there are 100 identical firms, each with the following technology: $f = K^{1/4}L^{1/2}$. Suppose also that in the short run $K = 1$, with $r = 1$, and $P = 4$. For the short run:

- If we assume that each firm is a profit maximizer, what is the problem that each firm solves?
- Calculate the MPL and each firm's demand for labor.
- Calculate each firm's profits.

(d) Calculate the market demand of labor.

Suppose that after considering basic activities required for life, there are 200 hours in a month which an individual can distribute between leisure and hours worked, that is $200 = h + l$ where h is leisure and l is labor. Suppose there are 1000 identical individuals, each with the same preferences: $U(h, x) = x^{1/2} + h^{1/2}$ over leisure and consumption x . Normalize the price of consumption x to 1.

(e) Suppose that an hour of labor is paid at the rate w per hour. Therefore, total consumption of x must equal the *labor income of the consumer*, i.e., $x = wl$. Use this information together with the time restriction $h + l = 200$ to state the individual's budget restriction in terms of x and h .

(f) State the individual's problem and the optimality conditions.

(g) If $w = 1$, find the optimum of leisure h and consumption x .

(h) Find the individual supply of l (do not assume $w = 1$).

(i) Find the market supply of labor.

Assume that the amount of labor (hours worked) is traded in a competitive market.

(j) What will be the equilibrium wage rate w^* ?

(k) What will be the equilibrium level of labor?